INCH-POUND
MIL-C-11015/23E
25 October 2021
SUPERSEDING
MIL-C-11015/23D
14 May 2001

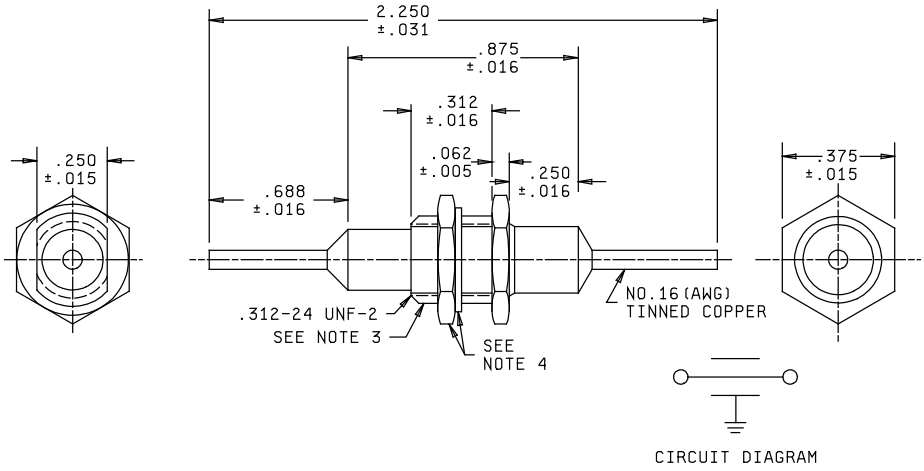
MILITARY SPECIFICATION SHEET

CAPACITOR, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE), (FEED-THRU), STYLE CK72

INACTIVE FOR NEW DESIGN AFTER 31 MARCH 1999.
FOR REPLACEMENT PURPOSES ONLY.

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-C-11015.



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in millimeters and are given for general information only.
3. The threaded portion shall extend to within .047 inch (1.19 mm) of the shoulder.
4. A nut and internal-tooth lockwasher shall be supplied with each capacitor when requested by the procuring activity.

Inches	mm
.005	0.13
.015	0.38
.016	0.41
.031	0.79
.062	1.57
.250	6.35
.312	7.92
.375	9.53
.688	17.48
.875	22.23
2.250	57.15

FIGURE 1. Dimensions and configuration.

AMSC N/A
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REQUIREMENTS

Dimensions and configuration: See figure 1.

Case type: Feed-thru.

Capacitance value: See table I.

Capacitance tolerance: ± 20 percent (M).

Rated temperature: -55°C to +85°C.

Dielectric withstanding voltage: In accordance with MIL-C-11015.

Dielectric (DWV):

Test voltage: 250 percent of rated voltage, applied between the case and the terminal.

Body insulation: Not applicable.

Barometric pressure (reduced): In accordance with MIL-C-11015 and MIL-STD-202-105, condition B (50,000 ft).

Test potential: 1,550 volts dc.

Insulation resistance (IR): In accordance with MIL-C-11015 and MIL-STD-202-302, condition B. 500,000 megohms, minimum, measured between either insulated point and capacitor case.

Dissipation factor (DF): 1.2 percent, maximum.

TABLE I. Style CK72 characteristics.

Part or identifying number (PIN)	Rated voltage (V_{dc})	Rated temperature and voltage-temperature limits	Capacitance (pF)	Capacitance tolerance
CK72AX101M	1,500	AX	100	M
CK72AX471M	1,500	AX	470	M
CK72AX681M	1,500	AX	680	M
CK72AX102M	1,500	AX	1,000	M

Vibration, high frequency: In accordance with MIL-C-11015 and MIL-STD-202-204, condition D (20 g's).

Thermal shock and immersion: In accordance with MIL-C-11015.

IR: 200,000 megohms, minimum.

Terminal strength: Not applicable.

Moisture resistance: In accordance with MIL-C-11015.

DWV: 250 percent of rated voltage.

IR: 150,000 megohms, minimum.

Cap.: Within tolerance of table I value.

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Solderability: In accordance with MIL-C-11015; 2 terminals.

Life (at elevated ambient temperature): In accordance with MIL-C-11015.

Test potential: 200 percent of rated voltage.

IR: 200,000 megohms, minimum (at +85°C), 300,000 megohms, minimum (at +25°C).

DF: 1.5 percent, maximum (at +25°C).

Marking: The operating temperature range and voltage-temperature limit, capacitance, and capacitance tolerance shall be indicated on the capacitor by color coding as shown on figure 2 and table II. The MIL identifier (yellow spot) shall be applied to the capacitor body diametrically opposite the color spots. At the option of the manufacturer, capacitors may be marked in accordance with MIL-C-11015. In either case, full marking shall appear on the unit package.

TABLE II. Color code.

Color	Operating temperature range and voltage-temperature limits	Nominal capacitance		Capacitance tolerance
		First and second significant figures	Multiplier 1/	
Black	---	0	1	M
Brown	---	1	10	---
Red	AX	2	100	---
Orange	---	3	1000	---
Yellow	---	4	---	---
Green	---	5	---	---
Blue	---	6	---	---
Purple (violet)	---	7	---	---
Gray	---	8	---	---
White	---	9	---	---

1/ The multiplier is the factor by which the two significant figures are multiplied to yield the nominal capacitance.

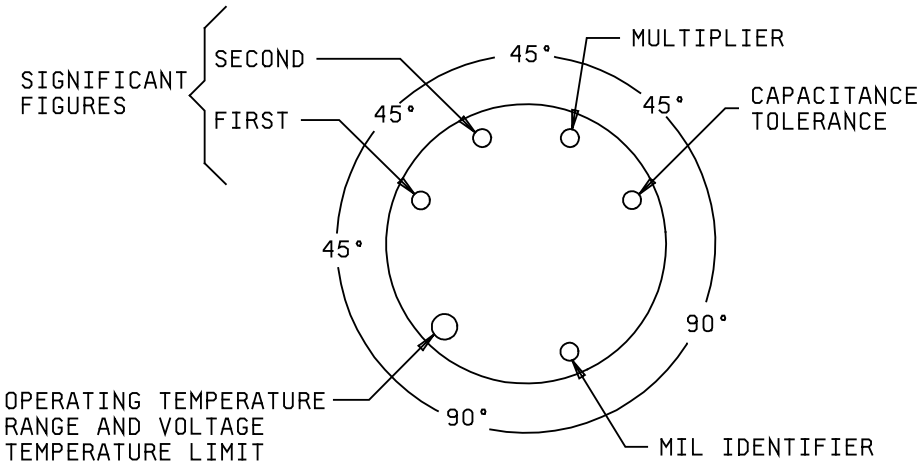


FIGURE 2. Color code marking.

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Referenced documents: In addition to MIL-C-11015, this document references the following:

MIL-STD-202-105
MIL-STD-202-204
MIL-STD-202-302

Changes from previous issue: The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
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Navy - EC
Air Force - 85
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Preparing activity:
DLA - CC

(Project 5910-2021-035)

Review activities:
Army - MI
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